SCS 1307
Probability & Statistics

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Binomial Distribution

Consider an experiment which has two possible outcomes, one which is considered as the 'success' and the other is 'failure'.

A Binomial situation arises when ***n*** independent trials of the experiment are conducted, for example

- Toss a coin 6 times, considering obtaining a head on a single trial as the success
- Roll a die 10 times, considering obtaining an even number as the 'success'

Let p be the probability that an event will happen in any single trial.

The probability that the event will fail to happen in any single trial is 1-p.

The probability that the event will happen exactly $\boldsymbol{k}$ times in $\boldsymbol{n}$ independent trials is given by the probability function.

$$Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

When the random variable X denote the number of successes in $\boldsymbol{n}$ independent trials, the above given function is often called the Binomial Distribution since for x = 0,1,.........,$\boldsymbol{n}$**,** it corresponds to successive terms in the binomial expansion.

Example: The probability of getting exactly 2 heads in 6 tosses of a fair coin is,

$X \sim Bin(6,1/2)$

$P(X=2) = {}^{6}C_{2}\,(1/2)^{2}\,(1/2)^{4} = 15/64$

Exercise

1) Find the probability that in tossing a fair coin three times there will appear

   (a) 3 heads

   (b) 2 tails and 1 head

   (c) at least 1 head

   (d) not more than 1 tail

Solution

1) X~Bin(3,1/2). Probability of getting

(a) 3 heads

(b) 2 tails and 1 head ;   Let Y be the number of tails :

Solution

1) Probability of getting

   (c) at least 1 head

   (d) not more than 1 tail (3 Heads or 2 Heads)

Exercise

Find the probability that in 5 tosses of a fair die a 3 appears (a) twice (b) at most once (c) at least two times.

Solution

The probability that in 5 tosses of a fair die a 3 appears

$X \sim Bin(5,1/6)$

(a) twice =

(b) at most once =

(c) at least two times =

Exercise

Find the probability that in a family of 4 children there will be

(a) at least 1 boy,

(b) at least 1 boy and at least 1 girl.

Assume that the probability of a male birth is 1/2.

Solution

The probability that in a family of 4 children there will be

Let X be a child being a boy: X~Bin(4,1/2)

(a) at least 1 boy =

(b) at least 1 boy and at least 1 girl =

Some properties of the Binomial Distribution

Mean = np

Variance =npq where q=1-p

Example

In 100 tosses of a fair coin what is the expected number of Heads ?

$E(X) =$

Exercise

If the probability of a defective bolt is 0.1, find

(a) the mean

(b) the standard deviation

for the number of defective bolts in a total of 400 bolts.

Solution

If the probability of a defective bolt is 0.1 and total number of bolts available is 400,

(a) the mean =

(a) the standard deviation =

Exercise

The random variable X has a binomial distribution with mean 5.76 and standard deviation 1.92. Find P(X=6)

Exercise

The random variable X has a binomial distribution with mean 5.76 and standard deviation 1.92. Find P(X=6)

$X \sim Bin(n,p)$

$np =$

$np(1-p)=$